SU$(2)_L$-Chiralität aus der Galois-Struktur:

Su/Sd-Trennung und Cl(6)-Gradparität

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Su/Sd Split and Cl(6) Grade Parity

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Status markers

[B] *Established* — secured by script.

Abstract

The SU$(2)_L$ chirality structure of the Standard Model follows algebraically from the Cl(6) grade parity established in Doc. 341. The pseudoscalar $\gamma_5 = e_1e_2e_3e_4e_5e_6$ has eigenvalue $(-1)^r$ on grade-r elements. The Su/Sd split (Doc. 344) is therefore exactly the right/left split: Su = even grades = right-handed **[B]**, Sd = odd grades = left-handed **[B]**. The open question from Docs. 347/348 is thereby closed **[B]**.

1 Theorem A — Grade parity and γ_5 eigenvalue [B]

$\gamma_5 \cdot X = (-1)^r X$ for grade-r element $X \in$ Cl(6). Even grades: $\gamma_5 = +1$ (right-handed). Odd grades: $\gamma_5 = -1$ (left-handed). **[B]**

2 Theorem B — Su = even grades = right-handed [B]

Su[0]=ν: grade 0 → right. Su[1]=$\bar{u}_d$: grade 2 → right. Su[2]=u: grade 4 → right. Su[3]=e_R: grade 6 → right. All $\gamma_5 = +1$. **[B]**

3 Theorem C — Sd = odd grades = left-handed [B]

Sd[0]=$\bar{\nu}$: grade 1 → left. Sd[1]=d: grade 3 → left. Sd[2]=$\bar{d}$: grade 5 → left. Sd[3]=e_L: grade 7 → left. All $\gamma_5 = -1$. **[B]**

4 Theorem D — $SU(2)_L$ doublets from Sd sector [B]

Left-handed doublets: $(\nu_L, e_L) = (\text{Sd}[0], \text{Sd}[3])$; $(u_L, d_L) = (\text{Sd-quarks})$. Right-handed singlets: $e_R = \text{Su}[3]$; $u_R = \text{Su}[2]$; $\nu_R = \text{Su}[0]$. $SU(2)_L$ acts on the Sd sector (left-handed) only. **[B]**

5 Theorem E — $\mathbb{Z}_3$ fixed points/orbit = chirality [B]

From Doc. 341: Vss$[0, 2, 4]$ (σ-fixed points) = right-handed; Vss$[1, 3, 5]$ (σ-three-orbit) = left-handed. The $SU(2)_L$ chirality split is a direct consequence of the $\mathbb{Z}_3$ symmetry of T^4/Z_3. **[B]**

6 Summary

Theorem	Content	Status
A	γ_5 eigenvalue $(-1)^r$ on grade r	**[B]**
B	Su = even grades = right-handed	**[B]**
C	Sd = odd grades = left-handed	**[B]**
D	$SU(2)_L$ doublets = Sd sector	**[B]**
E	$\mathbb{Z}_3$ fixed points/orbit = right/left	**[B]**

The $SU(2)_L$ chirality structure is now derived from the Galois/Clifford algebra **[B]**. Open: exact CKM mixing angles and CP phase δ_{CP} (Doc. 348) **[S]**.

Verification script

`pruef_349_chiralitaet.py` — Theorems A–E, all assertions passed **[B]**.

Note on AI assistance

Prepared with support from Claude (Anthropic). Physical interpretations: Johann Pascher (ORCID 0009-0000-6518-4064).

Bibliography

[1] J. Pascher, Doc. 341, FFGFT corpus (Sept. 2026).

[2] J. Pascher, Doc. 344, FFGFT corpus (Sept. 2026).

[3] J. Pascher, Doc. 347, FFGFT corpus (Sept. 2026).

[4] C. Furey, Ph.D. Thesis, Waterloo 2015, arXiv:1611.09182.